

Four Queens

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Wolfgang has a square chessboard of side length eight squares. He also has four chess pieces, all of which are identical queens (which can move vertically, horizontally and diagonally, but cannot jump over each other).

He has marked some squares of the chessboard, and he now wants to place the pieces so that all marked squares are *controlled*. A square is said to be *controlled* if it is occupied by a piece, or if a piece can move directly to it.

Help Wolfgang determine the number of ways to place his pieces to control all marked squares, and give an example of how to do it.

Note: swapping two queens does not constitute a new arrangement.

Input

The input consists of eight lines, representing the rows of the board. Each line consists of eight characters, representing the squares in that row.

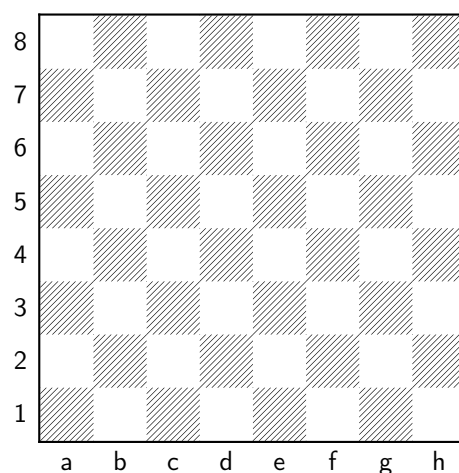
Each of these characters is either:

- '*' (representing a marked square), or
- '.' (representing an unmarked square).

Output

The first line of output should contain a single integer, the number of arrangements.

If the number of arrangements is at least one, the next four lines of output should detail any satisfactory arrangement, by listing the coordinates of the queens. The chessboard will be labelled as shown below:



Examples

standard input	standard output
<pre> *****. *****. *****. *****. *****. *****. *****. *.***** </pre>	<pre> 1 c3 d6 d7 g2 </pre>
<pre> ***** ***** ***** ***** ***** ***** ***** </pre>	<pre> 0 </pre>
<pre> .*****. *....*.. *....*.. *....*.. *...*... **..*... .*****.*.. </pre>	<pre> 11298 b4 c8 d6 f2 </pre>

Note

Figure 1

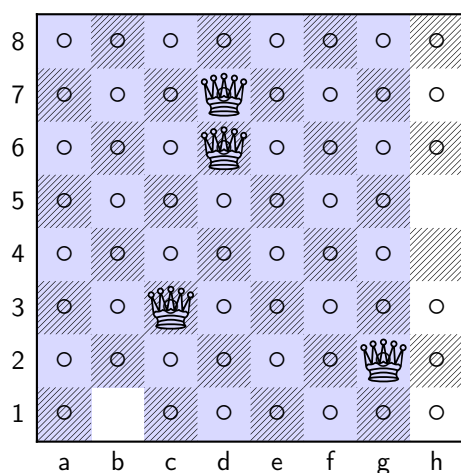
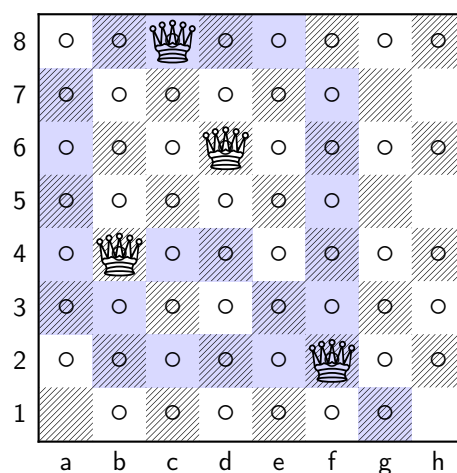


Figure 2



For the first sample case, Figure 1 shows the unique way of placing Wolfgang's four queens such that all marked squares are controlled.

For the second sample case, it can be shown that no placement of the four queens can control all marked squares.

For the third sample case, Figure 2 shows one of the possible arrangements of the four queens such that all marked squares are controlled.